

Instructor: Date:

Learner: Regional Branch:

Program: Center:

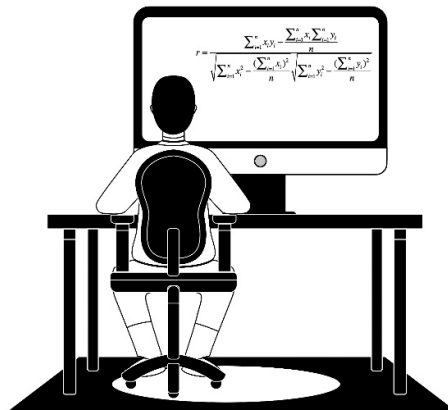
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Let's Start!

A Tool That Makes a Programmer's Job Easier



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Ask your partner:

- What are the benefits of working with "programmer friendly" languages?
- Why do you think that the syntax of some programming languages is similar to the human language?

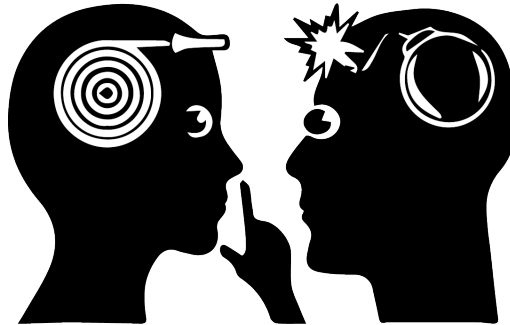
Can Do Statements:

- I can talk about automatic programming.
- I can give information about automatic programming.
- I can explain what automatic programming is and some related facts.



Let's Explore!

Why is it taking so long to write this subroutine?



Instructions: Read the text below.

- Alex:** That's it, I think my whole weekend is ruined!
- Jane:** What is it? You look very concerned.
- Alex:** Well, I am trying to write a subroutine, but it is taking too long. I guess I am working all weekend until I finish it.
- Jane:** Let me take a look. Is that the subroutine?
- Alex:** Yes, I am writing line after line of code, but apparently, it takes forever.
- Jane:** I think I know how to save your weekend. I guess you don't know about automatic programming.
- Alex:** What is automatic programming?
- Jane:** Automatic programming is a tool that software developers use in order to facilitate their work by writing codes that are able to write other codes.
- Alex:** I don't understand, is this some kind of magic?
- Jane:** Not really. Automatic programming is a method of computer coding in which software programmers write high-level applications that in turn generate software code based on artificial specifications.
- Alex:** That sounds fantastic! Are these specifications defined automatically?
- Jane:** No, the programmer specifies the automatic coding parameters. There are two types of automatic programming. First, we have generative programming that uses standard logical libraries. Second, we find source code generation, which uses models or templates based on mathematical algorithms.
- Alex:** I am using automatic programming to solve the subroutine. I am saving a lot of time. Finally, Jane, do you know how I can express these algorithms? They are recurrent in the entire program.
- Jane:** Are those repeated over and over?
- Alex:** Sure!
- Jane:** You can use a programming idiom to do it. Programming idioms are expressions used for performing not only recurrent tasks but also for more complex algorithms, and even full data structures or patterns.
- Alex:** Thanks, Jane. You are very helpful.



Let's Practice!

Why is automatic programming important?

Instructions: Select all the correct answers.

1. What does automatic programming help programmers with?
 - a. It helps them write high-level applications.
 - b. It helps them turn the computer on.
 - c. It helps them generate software code on artificial specifications.
 - d. It helps them use standard logical libraries to produce code.



Let's Practice!

General Information About Automatic Programming

Instructions: Select the correct answer.

1. Automatic programming is a tool for software developers. Is ____ tool capable of making their job easier?
 - a. this
 - b. these
 - c. those
 - d. there
2. There are two types of automatic programming, generative programming and source code generation. Are ____ useful when working with complex apps?
 - a. this
 - b. that
 - c. these
 - d. there
3. People use programming idioms to perform various tasks. Are _____ able to handle high-level specifications?
 - a. that
 - b. there
 - c. those
 - d. this





Extension Activity

Automatic programming

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about automatic programming.
- List the most remarkable characteristics about it.
- Make a poster in which you present these characteristics.
- Share your poster with the class.

Self-Assessment

- I find information about automatic programming.
- I list the characteristics of automatic programming.
- I share my work with the class.

Yes

No

☐ ☐
☐ ☐
☐ ☐


Target Words

- **automatically** (adverb): performed without human intervention
e.g. Are these specifications defined automatically?
- **template** (noun): something that works as pattern for creation
e.g. Second, we find source code generation, which uses models or templates based on mathematical algorithms.
- **generative** (adjective): having the ability to originate or reproduce
e.g. First, we have generative programming that uses standard logical libraries to produce code.
- **concerned** (adjective): to be worried or upset
e.g. What is it? You look very concerned.
- **facilitate** (verb): to make a task easier
e.g. Automatic programming is a tool that software developers use in order to facilitate their work by writing code that writes code.
- **specify** (verb): to mention clearly and precisely
e.g. The programmer specifies the automatic coding parameters.



Learning Points

Demonstrative Pronouns in Questions

We use **demonstrative pronouns** to replace a noun. We can use demonstrative pronouns in questions, and it is important that the noun we replaced for the demonstrative pronoun is clear for our interlocutor.

Demonstrative pronouns are words we use to point at something specific inside a sentence. They can be singular or plural, and refer to objects close or far in space or time.

There are four demonstrative pronouns: *this, these, that and those*.

Examples: ***This is*** the new script for the application.

He is correcting ***these*** code errors.

That software was not ideal for our needs.

Those are the corrections that Sharon wrote.

Demonstrative pronouns for asking questions:

Before you use demonstrative pronouns to ask questions, it is very important that the person we are talking to knows in advance the thing or things the demonstrative pronoun is replacing.

Grammatical structure:

Auxiliary verb + demonstrative pronoun + complement + ?

Example: ***Are those*** programs safe?

Do these programs offer everything we need?

	Close in space and time	Far in space and time
Singular	This	That
Plural	These	Those

Examples

- *Is that the subroutine?*
- *Is this some kind of magic?*
- *Are these specifications defined automatically?*
- *Are those repeated over and over?*





Learning Points

Adjectives Used to Describe Automatic Programming

Automatic programming is a tool that software developers use in order to facilitate their work. You can use some specific adjectives to describe automatic programming e.g. high-level, artificial, mathematical, logical, etc.

Let's see those adjectives in the next text:

Automatic programming is a method of computer coding in which software programmers write **high-level** applications that in turn generate software code based on **artificial** specifications. In some way, translators are automatic programs that use a **high-level** language that they translate into a **lower-level** language, which is the specification.

There are two types of automatic programming. First, we have generative programming that uses standard **logical** libraries for improving programming speed and efficiency. This process saves time since the programmer doesn't need to re-implement certain automatic elements during compile. Second, we find source code generation, which uses models or templates based on **mathematical** algorithms. Once the model is created, a source code generator creates the code based on how the components connect to each other in the resulting program. Automatic programming is also referred to as metaprogramming.

Adjective	Meaning
High-level	Something very detailed, with a strong level of complexity.
Low-level	Something not very detailed, with a mild level of complexity.
Artificial	Something that humans created.
Mathematical	Something exact, precise, and related to mathematics.
Logical	Something that follows the principles of logic.

Examples

- *Automatic programming is a method of computer coding in which software programmers write high-level applications that in turn generate software code based on artificial specifications.*
- *We find source code generation, which uses models or templates based on mathematical algorithms.*





Learning Points

What does programming idiom mean?

Similar to natural language, programming languages have expressions that we often repeat. We can use those expressions in certain circumstances.

We use a programming idiom when we want to express a recurrent concept in one or several programming languages. In general, a programming idiom is a way of expressing a repeated task that is a feature normally incorporated in the programming language that we are using. Programming idioms can even be present in different languages and have equivalent expressions and effects.

Programming idioms are expressions used for performing not only recurrent tasks, but also more complex algorithms, and even full data structures or patterns.

Like learning a natural language, remembering the idioms associated with a programming language, and how to use them in the appropriate context, is an important part of gaining fluency in that language.

